Layer 6: Operational Services & Background Monitoring

monitoring_scheduler.py
Market Hours Daemon
(09:15-15:30 IST)

notification_service.py
Drift Alerts (>= 5%)
In-App & Email Notices

diagnosis/service.py
Demat CAS PDF Parser
HHI Concentration Score

rebalancing_service.py
Before/After Preview
Execution Order Delta

Drift Alert >= 5%

Layer 1: Client Presentation Layer (Next.js 16.4.2 & React 19.3.0 UI)

Stress Tests & AI
ConversationalAssistant
What-If Crash Scenarios

Why This Portfolio?
StockWhyModal.tsx
Retail Thesis & Factors

Demat Diagnosis
DiagnosisView.tsx
CAS PDF Health Audit

Portfolio Generator
PortfolioGenerator.tsx
Risk Questionnaire

Rebalance Preview
RebalancePreviewModal
Taxes, STT & Lot Orders

My Portfolio
HoldingsView.tsx
Live NAV & Drift Radar

Layer 2: API Gateway & Security Layer (FastAPI 0.115 Backend)

Security & Rate Limits
Argon2id Hasher & JWT Bearer Tokens
Sliding-Window Rate Limiter
(middleware/)

Investor & Market Routers
api/v1/auth.py & profile.py
api/v1/risk_assessments.py &
market_data.py

Portfolio & Operational Routers
api/v1/portfolios.py & rebalancing.py
api/v1/diagnosis.py & notifications.py

Layer 3: Orchestration & State Management (LangGraph 0.4)

StateGraph Orchestrator
backend/agent/graph.py
Dynamic Agent Flow

AgentState Dataclass
backend/agent/state.py
Context & Shared Memory

Fiduciary Gate Loops
Critic Rejection &
Rework Self-Correction

Execution Telemetry
agent_execution_runs
Token & Latency Tracing

Layer 4: Multi-Agent AI & Intelligence Layer (13 LangGraph Nodes)

1. Coordination & Suitability

manager_node
Coordinates Workflow & Subtasks

suitability_check_node
Enforces SEBI Suitability Rules

2. Specialized Research (Gemini 3.8 Flash)

macro_analyst_node: RBI Repo & Inflation

market_analyst_node: Nifty 50 Trend & VIX

fundamental_analyst_node: ROE >= 15%, D/E

news_analyst_node: Corporate Sentiment

3. Strategy, Risk Critique & Persistence

analyst_node: Lead Strategist Allocation

critic_node: Adversarial Fiduciary Critic

moderator_explanation_node: Stock-Why

persistence_node: Atomic PostgreSQL Commit

Layer 5: Quantitative & Mathematical Engine (SciPy & NumPy)

optimizer.py
SciPy SLSQP Solver
Max Sharpe / Min Risk

allocation_rounding.py
Whole-Share Floor Rounding
Discrete Lot Sizing

Cash Conservation
Preserves Unallocated Cash
Exact Budget Matching

transaction_costs.py
Indian Statutory Frictions
STT, Stamp Duty, GST

Layer 8: External Integrations & Market Data Feeds

NSE Official Archive
EQUITY_L.csv
(2,639 Listed Equities)

BSE Directory
Dual-Listed Scrip
Matching
Universal ISIN Keys

Yahoo Finance API
Daily OHLCV Historical
Bars
Corporate Action Splits

Reserve Bank of India
Official Policy Bulletins
Repo Rate & Inflation

Google Gemini API
Gemini 3.8 Flash
Multi-Agent Committee

Layer 7: Relational Data Persistence (PostgreSQL 16 - 62 Tables Master)

Users & Security
app_users, user_profiles
risk_assessments, goals

Securities & Prices
securities (2,639 Listed)
market_prices,
fundamentals

Portfolios & Decisions
investor_portfolios,
versions
portfolio_security_decisions

Telemetry & Alerts
agent_execution_runs
explanations, notifications